NEXMUTATION

Enterprise Architecture & Hosting Manual
Domain: `nexmutation.com` | Registered at: GoDaddy
Date: October 2024 | Version: 1.0 (Production)

NexMutation Website: Deployment, Hosting & Content Management Guide

This comprehensive document outlines the architectural hosting options, DNS configurations, and step-by-step procedures for deploying the **NexMutation** website, hosting technical YouTube video showcases, publishing new research blogs, and connecting to your existing GoDaddy domain.

> **Important Domain Notice**
> Your domain **nexmutation.com** is already registered with **GoDaddy**. You do **NOT** need to buy a domain. You only need to choose a hosting provider and point your GoDaddy DNS records to it.

1. Hosting Platform Comparison

Platform	Cost	Global Edge CDN	SSL (HTTPS)	Video & Blog Handling	Best For
Vercel	**100% Free**	Ultra-Fast (300+ Edge Nodes)	Automatic Free SSL	Unlimited YouTube embeds + Static Blogs	Top Tech & AI Startups (Recommended)
Netlify	**100% Free**	Global High-Speed CDN	Automatic Free SSL	Supports Free Decap CMS Visual Dashboard	Non-technical visual blog writing
Cloudflare Pages	**100% Free**	Fastest Anycast Network	Automatic Free SSL	Unlimited bandwidth & requests	Heavy international traffic
Firebase Hosting	**100% Free Tier**	Google Cloud Global CDN	Automatic Free SSL	Static files + Google Cloud backend	Google Cloud / Firebase users
GoDaddy cPanel / Hostinger	**$2.50 – $7/mo**	Single Server (No edge CDN)	May cost extra on GoDaddy	Full PHP / MySQL Server	Traditional LAMP / WordPress hosting

2. Method 1: Deploying to Vercel (Recommended - 100% Free)

Vercel is the hosting standard for modern web applications and AI companies (used by OpenAI, Nike, and Next.js).

1. **Create a Free Account**: Go to vercel.com and sign up with GitHub, GitLab, or Email.
2. **Option A (Via GitHub - Recommended for Continuous Deployments)**:
   - Push your `NexMutation_v1` folder to a private or public GitHub repository.

- In Vercel dashboard, click **"Add New Project"** and select your GitHub repository.
- Framework Preset: **Other** (Static). Root Directory: `./`.
- Click **Deploy**. In 30 seconds, your site is live on a secure URL (e.g. `nexmutation.vercel.app`).

3. **Option B (Direct Upload via CLI)**:

```
npm i -g vercel
cd /home/gaian/Desktop/Nex/NexMutation_v1
vercel --prod
```

3. Method 2: Deploying to Netlify (Free Visual CMS Support)

Netlify provides identical enterprise-grade speed and enables an optional **Visual Admin Login** at `/admin` to write blogs without code.

1. Sign up for a free account at netlify.com.
2. Drag and drop the `NexMutation_v1` folder directly into the Netlify "Sites" area.
3. Netlify provisions a free high-speed CDN and SSL certificate instantly.

4. Connecting GoDaddy Domain (nexmutation.com) to Your Host

Once deployed on Vercel or Netlify, configure your GoDaddy DNS settings:

1. Log in to your **GoDaddy Account** > **My Products** > **nexmutation.com**.
2. Click **Manage DNS**.
3. Update or add the following records:

Hosting Provider	Type	Name	Value / Destination	TTL
Vercel (Root Domain)	`A`	`@`	`76.76.21.21`	1 Hour / Default
Vercel (WWW Subdomain)	`CNAME`	`www`	`cname.vercel-dns.com`	1 Hour / Default
Netlify (Root Domain)	`A`	`@`	`75.2.60.5`	1 Hour / Default
Netlify (WWW Subdomain)	`CNAME`	`www`	`[your-site-name].netlify.app`	1 Hour / Default

> **DNS Propagation Time**
>
> DNS updates usually take **5 to 30 minutes** to propagate globally (maximum 24 hours). Both Vercel and Netlify will automatically generate and renew a free Let's Encrypt SSL certificate for `https://nexmutation.com`.

5. How to Add New YouTube Tech Videos to the Website

Because YouTube hosts the video, you never pay server storage costs. To add a new tech video showcase:

Step 1: In `portfolio.html` (or `index.html`)

Add a new card with a GLightbox video link:

```
<div class="col-lg-4 col-md-6 portfolio-item filter-app">
  <div class="portfolio-wrap">
    <img src="https://img.youtube.com/vi/YOUR_VIDEO_ID/maxresdefault.jpg" class="img-fluid" alt="T
ech Tutorial Title">
    <div class="portfolio-info">
      <h4>Building Enterprise RAG Systems</h4>
      <p>AI Architecture Walkthrough</p>
      <div class="portfolio-links">
        <a href="https://www.youtube.com/watch?v=YOUR_VIDEO_ID" class="glightbox pulsating-play-bt
n" title="Watch Video">
          <i class="bi bi-play-circle"></i>
        </a>
      </div>
    </div>
  </div>
</div>
```

Replace `YOUR_VIDEO_ID` with your YouTube video code. The lightbox automatically opens it in an overlay player.

6. How to Publish New Blog Posts

1. **Duplicate Template**: Copy `blog-details.html` and name it with your topic, e.g. `blog-llm-fine-tuning.html`.
2. **Edit Article Content**:
   - Update `<title>` and `<meta name="description">` in the `<head>`.
   - Replace the article title, author name (Ashish Singh or Neha Kumari), publication date, and body paragraphs.
3. **Add to `blog.html` Listing**:

   Add a summary card to the grid in `blog.html`:

```
<article class="blog-item col-lg-4">
  <div class="post-img">
    <img src="assets/img/blog/blog-7.jpg" alt="Article Cover" class="img-fluid">
  </div>
  <p class="post-category">LLM Engineering</p>
  <h2 class="title"><a href="blog-llm-fine-tuning.html">Fine-Tuning Llama 3 for Enterprise C
ompliance</a></h2>
  <div class="d-flex align-items-center">
    <div class="post-meta">
      <p class="post-author">Ashish Singh</p>
      <p class="post-date"><time datetime="2024-11-01">Nov 1, 2024</time></p>
    </div>
  </div>
</article>
```

4. **Update `sitemap.xml`**: Add the new URL so search engines immediately index it.

7. Design & Theme Reference Cheat Sheet

The website is built with a dual-theme system (Light & Dark mode) controllable via the header toggle or CSS variables in `assets/css/main.css`.

Variable / Element	Light Theme (Deep Tech)	Dark Theme (Cyber Obsidian)	Usage
`--background-color`	`#ffffff` / `#f8fafc`	`#080e1a` / `#0c1424`	Page and section background
`--surface-color`	`#ffffff`	`#0f172a`	Cards, panels, widgets
`--heading-color`	`#0f172a` (Slate 900)	`#f8fafc` (White)	H1, H2, H3 typography
`--default-color`	`#334155` (Slate 700)	`#94a3b8` (Slate 400)	Paragraphs & body text
`--accent-color`	`#0284c7` (Electric Azure)	`#38bdf8` (Cyan Sky)	Buttons, badges, highlights
`Header Scrolled`	`rgba(255, 255, 255, 0.95)`	`rgba(8, 14, 26, 0.95)`	Frosted glass sticky header

8. Recommended Next Steps for Launch

1. **Push to GitHub**: Create a repository (e.g. `NexMutation_Website`) and push this directory.
2. **Connect to Vercel or Netlify**: Import the repository with 1 click.
3. **Update GoDaddy DNS**: Point the `A` record to Vercel/Netlify as specified in Section 4.
4. **Verify SSL**: Your website will securely load at `https://nexmutation.com` worldwide within 30 minutes.

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